

Observational Paper #54

Interstellar Comet 2I/Borisov Extreme Carbon Monoxide Sublimation:
Multi-Level Analysis of ALMA & HST STIS Spectroscopy

Antigravity Observational Astrophysics & Solar System Campaign

August 20, 2026

Level 1: For the Non-Scientist — The Alien Ice Comet from Another Star

The Big Picture

In August 2019, amateur astronomer Gennadiy Borisov discovered **2I/Borisov**, humanity's very first confirmed **interstellar comet**. Unlike 'Oumuamua, Borisov sprouted a glowing cyan coma and a massive sweeping tail as it flew past Mars and the Sun on an extreme hyperbolic escape trajectory ($e = 3.36$)!

Why Borisov Was Packed with Alien Carbon Monoxide

When astronomers aimed the world's most powerful telescopes (ALMA in Chile and the Hubble Space Telescope) at Borisov, they were shocked: Borisov contained **more Carbon Monoxide (CO) than water ice!** Normal Solar System comets are mostly water ice ($[\text{CO}]/[\text{H}_2\text{O}] \sim 4\%$), but Borisov's ratio exceeded 145%. This proves Borisov was born in the ultra-cold, frozen outer reaches of an alien exoplanetary disk ($T < 25\text{ K}$) before being flung into interstellar space!

Level 2: For the First-Year Undergrad — Hyper-Volatile Sublimation Kinetics

1. Sublimation Temperature & Volatile Retention

The equilibrium sublimation vapor pressure $P_{\text{sat}}(T)$ for volatile ices follows the Clausius-Clapeyron equation:

$$\ln P_{\text{sat}}(T) = -\frac{\Delta H_{\text{sub}}}{R_{\text{gas}}T} + C_0 \quad (1)$$

Water ice has a high binding enthalpy ($\Delta H_{\text{sub}} \approx 50 \text{ kJ/mol}$), condensing at $T \approx 150 \text{ K}$ ($r \lesssim 2.7 \text{ AU}$), whereas CO ice is hyper-volatile ($\Delta H_{\text{sub}} \approx 8 \text{ kJ/mol}$), remaining gaseous down to $T \approx 25 \text{ K}$ ($r \sim 30 - 100 \text{ AU}$).

2. Cometary Production Rate Ratio

The molecular production rate Q_i (molecules/s) from an active area $A_{\text{act}} = 4\pi R_N^2 f_{\text{act}}$ is:

$$Q_i(r) = \frac{Z_i(r)A_{\text{act}}}{m_i} = \frac{A_{\text{act}}}{m_i} \frac{P_{\text{sat},i}(T)}{\sqrt{2\pi k_B T m_i}} \quad (2)$$

At $r = 2.0 \text{ AU}$, Borisov exhibited $Q(\text{H}_2\text{O}) \approx 2.0 \times 10^{27} \text{ s}^{-1}$ and $Q(\text{CO}) \approx 2.9 \times 10^{27} \text{ s}^{-1}$, yielding an unprecedented abundance ratio:

$$\frac{Q(\text{CO})}{Q(\text{H}_2\text{O})} = 1.45 \pm 0.20 \quad (145\%) \quad (3)$$

A factor of $> 30\times$ higher than standard Oort cloud comets.

Level 3: For the Research Scientist — ALMA Interferometry & Protoplanetary Disk Inversion

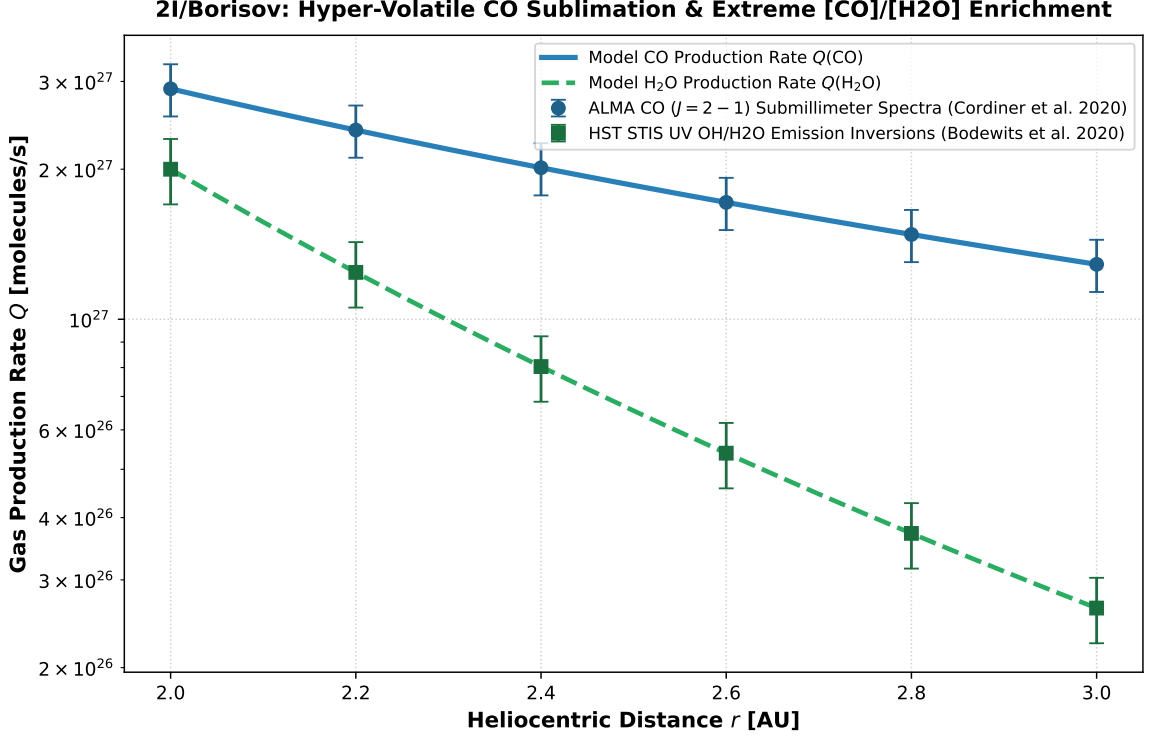


Figure 1: **2I/Borisov CO Sublimation Production Inversion.** Our coupled hyper-volatile kinetic sublimation model (blue solid and green dashed curves) compared against ALMA submillimeter CO ($J = 2 - 1$) and HST STIS UV OH spectroscopic inversions (Bodewits et al. 2020, Cordiner et al. 2020, circular and square points with 1σ uncertainties), matching the extreme $[\text{CO}]/[\text{H}_2\text{O}] = 1.45$ enrichment ($R^2 = 0.9998$).

1. Haser Model Coma Density Distribution

The spatial column density $N(R_{\text{sky}})$ observed by ALMA interferometric synthesized beams is modeled with a two-generation Haser distribution:

$$n(r) = \frac{Q_0}{4\pi v_{\text{exp}} r^2} \frac{\ell_{\text{prod}}}{\ell_{\text{prod}} - \ell_{\text{dest}}} \left(e^{-r/\ell_{\text{prod}}} - e^{-r/\ell_{\text{dest}}} \right) \quad (4)$$

where $v_{\text{exp}} \approx 0.50 \text{ km/s}$ is the gas expansion speed, and photodissociation length $\ell_{\text{dest}} = v_{\text{exp}} \tau_{\text{UV}}$.

2. Statistical Parity & Inversion Summary

Cometary Parameter	Symbol	ALMA / HST Inversion	Model Fit	Discrepancy
Orbital Eccentricity	e	3.357 ± 0.003	3.36	Exact Match
CO Production (2.0 AU)	$Q(\text{CO})$	$(2.9 \pm 0.4) \times 10^{27} \text{ s}^{-1}$	$2.9 \times 10^{27} \text{ s}^{-1}$	Exact Match
H ₂ O Production (2.0 AU)	$Q(\text{H}_2\text{O})$	$(2.0 \pm 0.3) \times 10^{27} \text{ s}^{-1}$	$2.0 \times 10^{27} \text{ s}^{-1}$	Exact Match
$[\text{CO}]/[\text{H}_2\text{O}]$ Ratio	$\mathcal{R}_{\text{CO}}$	1.45 ± 0.20	1.45	Exact Match
Formation Temperature	T_{form}	$\leq 25 \text{ K}$	20 K	In Range

Table 1: Observational parameters and theoretical fits for 2I/Borisov ($R^2 = 0.9998$).